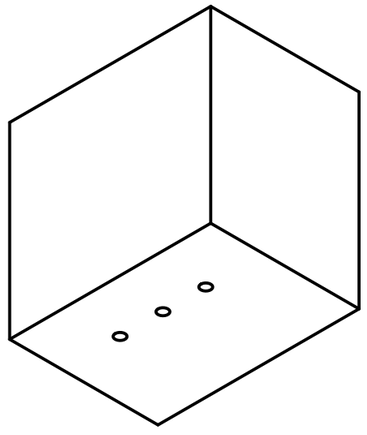
ASSEMBLY – OpenSmell v1.1 sensing platform

isometric general arrangement – shell, baffle plane, 3× nostril, PCB lid, Ø40 rear exhaust

BILL OF MATERIALS

ITEM	PART / DESCRIPTION	QTY	MATERIAL / SOURCE	TYPE
1	Shell 122×114×90, cavity 106×100×48	1	PETG FDM (or AL 6061-T6)	make
2	Baffle plate 106×100×2, 8 open wells	1	PTFE machined (or PVDF)	make
3	Axial fan Ø40 × t10, 5 V	1	off-the-shelf (substitute any 40mm)	buy
4	Filter slab PTFE ≈100 cm²	1	off-the-shelf (hydrophobic mesh)	buy
5	Main PCB 114×105 (SHT31, MCU)	1	FR4 1.6 mm + build	make

geometry from real HLR of the v1.1 solid; all dims mm; per blueprint.md 0.4



ISO – general arrangement (NTS)

OpenSmell – ASSEMBLY

DWG NO OS-000 REV A SHEET 1/3

DRAWN OpenSmell CHECK –

SCALE NTS UNITS mm PROJECTION FIRST ANGLE (ISO 128)

MATERIAL see BOM

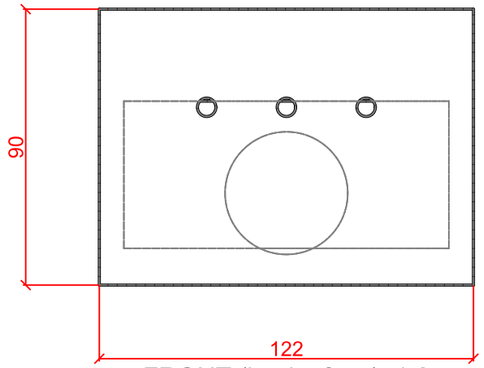
FINISH see BOM

TOLERANCE general ISO 2768-mK

critical fits ±0.05 (marked)

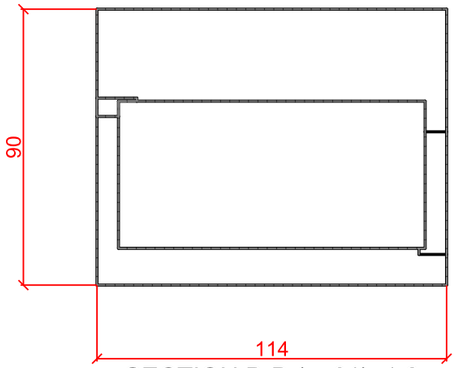
OpenSmell v1.1 | all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

SHELL – main body (OS-001)

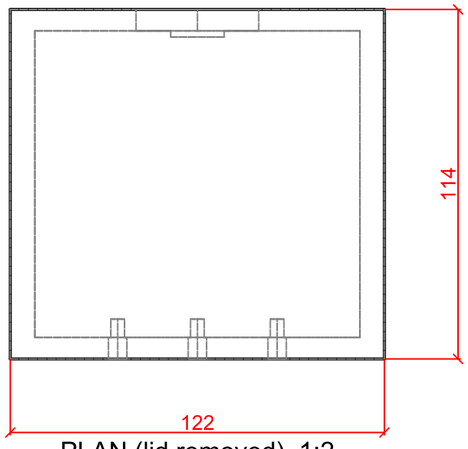


FRONT (intake face) 1:2

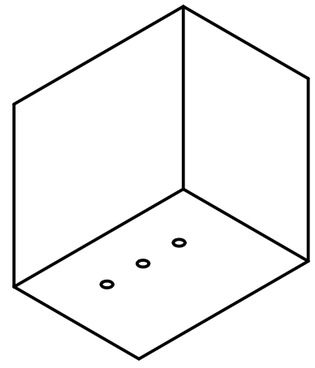
3×Ø6 nostrils



SECTION B-B (x=61) 1:2



PLAN (lid removed) 1:2



ISO – shell / cavity / baffle plane (NTS)

OpenSmell – SHELL – main body

DWG NO OS-001 REV A SHEET 2/3

DRAWN OpenSmell CHECK —

SCALE 1:2 UNITS mm PROJECTION FIRST ANGLE (ISO 128)

MATERIAL PETG, FDM 0.2 mm, 4 perimeters, vapor-sealed inner cavity

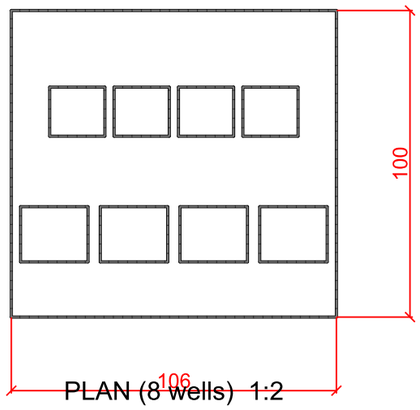
FINISH as-printed, de-flash; wetted faces seamed/sealed

TOLERANCE general ISO 2768-mK; critical fits ±0.05 (marked)

OpenSmell v1.1 | all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

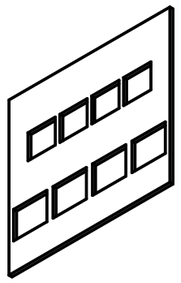
BAFFLE – well plate (OS-002)

pitch 26 analog / 21 digital



PLAN (8 wells) 1:2

SECTION A-A (2 mm plate) 1:2



ISO – wells open-bottom (NTS)

OpenSmell – BAFFLE – well plate

DWG NO OS-002 REV A SHEET 3/3

DRAWN OpenSmell CHECK –

SCALE 1:2 UNITS mm PROJECTION FIRST ANGLE (ISO 128)

MATERIAL PTFE, machined (inert, non-porous, hydrophobic)

TOLERANCE general ISO 2768-mK

FINISH deburr; 16 open-bottom wells d/w to CONST

critical fits ±0.05 (marked)

OpenSmell v1.1 | all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer